SIW Bandpass Filter — Project State

2.87 GHz dual-mode SIW filter on RO4350B

2026-09-03

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The goal

A Substrate Integrated Waveguide (SIW) bandpass filter:

Parameter	Target
Centre frequency	2.87 GHz
Bandwidth	143.5 MHz (5 % FBW)
Return loss	at most −16 dB in band
Response	Chebyshev, 0.1 dB ripple
Substrate	Rogers RO4350B, 0.76 mm, 0.5 oz copper (17.5 µm)
Vias	Ø1.5 mm, 3.0 mm pitch
Connector	Taoglas EMPCB.SMAFSTJ.B.HT, CPWG launch

For what to do next, see `PLAN.md`. Superseded documents are in `docs/archive/`.

> **2026-09-04 update — the board being fabricated is cascade N=3, not the dual-mode N=2 below.** Everything from here to "Reproducing this" is the dual-mode analysis exactly as measured on 2026-09-03 and is kept as real, correct physics — the bandwidth/match trade-off it documents is a genuine property of that topology and matters for the paper. But the fabrication decision it led to was **revised later the same session**: cascade N=3 is closer to spec on every axis (−13.37 dB / 189 MHz vs −10.76 dB / 316.6 MHz, at ~1.5× the board length) and was chosen instead. All fabrication outputs — CST macro, KiCad gerbers/drill/STEP, a 29-check `FAB_CHECKLIST.md`, a real-copper measurement, a PEC regression re-run that reproduced the golden to every digit, and the Taoglas datasheet verification — were built for **cascade N=3**, in `golden/cascade_N3/`. `siw kicad --topology cascade --N 3 --params ... --launch cpwg` reproduces all of it. `siw kicad --topology dualmode ...` refuses (dualmode has no `_geometry()` for CST/KiCad export) — that gap is real, not yet closed.

The design

Dual-mode N=2 — a single square SIW cavity carrying two degenerate modes, split by an internal floating iris row.

```
a = 57.158   w_iris = 15.304   x_iris = 28.685
y_wall = 9.849   w_ext = 16.352 mm        board 119.0 mm
```

	measured (PEC)	with real copper	target
Centre frequency	2.904 GHz	2.903	2.87
Return loss (worst in band)	−10.76 dB	−10.55	−16
−3 dB bandwidth	316.6 MHz	317.5	143.5
Insertion loss	−1.45 dB	−1.77	—
Reflection zeros	2 (2.787 / 2.897 GHz)	2	2

Rounding every dimension to 0.1 mm PCB resolution costs only 0.5 dB (−10.25 dB, 319 MHz), so the design is not perched on a knife edge.

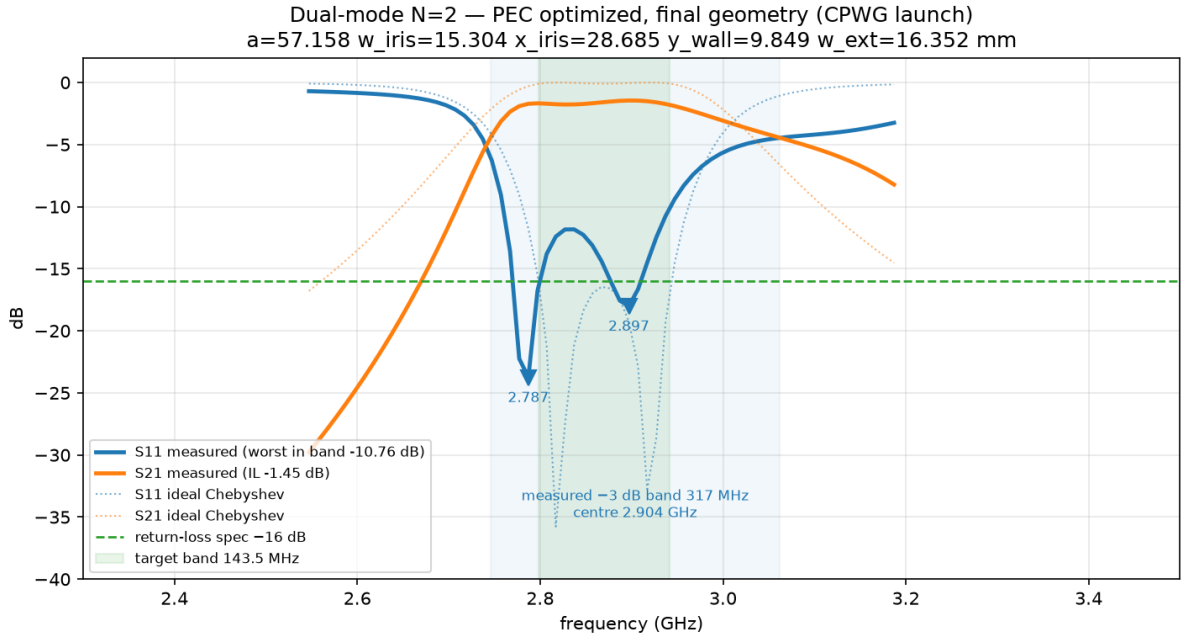


Figure 1: The final geometry against the ideal Chebyshev target. The response is structurally correct — two reflection zeros, one clean passband, low loss — but the nulls stop around −10 dB where the specification needs −16, and both zeros sit below the band centre.

The central finding: bandwidth and match are in opposition

The specification asks for a 143.5 MHz band **and** −16 dB return loss. On this topology those two are in direct conflict, and that is a measured result, not a tuning failure.

Four one-dimensional scans on the **assembled filter** (not a proxy structure), every point fine-mesh and verified:

`w_iris` at w_ext = 15.5 mm — the mode splitting

w_iris (mm)	11.0	12.0	13.0	14.0
bandwidth (MHz)	**145.6**	164.1	200.1	237.0
return loss (dB)	−1.75	−2.09	−3.36	−7.06

`w_ext` at w_iris = 15.26 mm — the launch aperture

w_ext (mm)	11.0	12.5	14.0	15.5	16.31
bandwidth (MHz)	35.0	65.6	122.1	284.4	310
return loss (dB)	−0.75	−1.18	−2.81	−6.86	**−10.35**
passband lobes	2	2	2	1	1

`w_ext` at w_iris = 11 mm: 120.7 / 127.7 / 145.6 / 166.4 / 366.5 MHz for 13 / 14.5 / 15.5 / 16 / 17.5 mm, return loss never better than −2.20 dB.

`y_wall` at w_iris = 11 mm: 147.1 / 146.7 / 153.6 / 169.1 MHz for 7.5 / 9.5 / 11 / 13 mm, return loss −1.77 to −2.27 dB — nearly inert.

Bandwidth alone is solvable. At `w_iris = 11 mm` the filter measures 145.6 MHz against a 143.5 MHz target — FBW 4.94 % against 5 %. The band was never "stuck"; it spans 35–366 MHz across the geometries scanned. What cannot be done is reaching that bandwidth *and* a usable match: return loss improves monotonically in whichever direction widens the band, on every parameter, with no interior optimum.

The project has chosen the good-match end (2026-09-03). The best return loss this structure produces anywhere is −10.76 dB.

A 30-evaluation local Bayesian search over a 5-dimensional ±2 mm box **confirmed** this geometry rather than improving it: neighbouring probes came back worse, the search spent its last evaluations resampling one point, and the 0.41 dB apparent gain over the previous best sits inside the measured coarse-mesh noise floor (re-sampling one geometry within 0.2 mm moved it 0.64 dB).

What the fabrication stack costs

Real copper: −0.32 dB insertion loss, −0.21 dB return loss. Centre frequency, bandwidth and pole structure all unchanged. This is the one trustworthy fabrication number, measured at the final geometry with the exterior ground planes modelled as lossy copper via a surface-impedance boundary condition.

Soldermask, silkscreen and ENIG cannot be modelled at this mesh density and are excluded. This was established by a one-factor-at-a-time ladder, each cell adding exactly one ingredient to the idealised baseline:

ingredient	thickness	ΔIL vs PEC	return loss	reflection zeros
copper (boundary condition, no geometry)	—	−0.32	−10.55	2 (ok)
soldermask	20 µm	−2.54	−0.98	3
silkscreen	12 µm	−4.02	−0.54	0
ENIG	4.5 µm	−7.19	−0.18	0

The damage scales monotonically with **thinness**, across two dielectrics and a conductor whose material properties have nothing in common. No physical mechanism does that; unresolved sliver elements do. Each layer spans the whole board inside a mesh whose finest cell is 200 µm, and adding one raises the degree-of-freedom count by only about 3 % — they are represented by a handful of cells with aspect ratios of 50:1 to 270:1. Resolving a 4.5 µm layer honestly would need roughly 1 µm cells, a thousand-fold local refinement.

The numbers in that table are therefore **not** fabrication penalties. They are the signature of a model outside its validity, and they are kept in the figure precisely so nobody re-adds those layers expecting meaning.

Consequences for the fabrication decisions

Choice	Decision	Basis
Finish	ENIG	board is unsoldered and exposed; gold/nickel is a real corrosion barrier. RF effect negligible — the plating sits on the *air* side of the ground planes, so cavity fields see copper first
Soldermask over the feed	**open it**	engineering judgement, **not** simulation: a 20 µm eps_r 3.8 film across the 0.2512 mm coplanar gap perturbs a launch that is already the weakest part of the design
Silkscreen	free choice	not measurable here; physically negligible

Choice	Decision	Basis
Vias tented vs open	**unanswerable**	the hollow-barrel geometry only exists together with a volumetric conductor model that is itself unresolvable

The earlier pad-only-launch fabrication study is **superseded** — its cells ran through the same unresolved-layer path. Its one figure that survives scrutiny is "+0.37 dB from real copper", close to today's properly measured 0.32 dB.

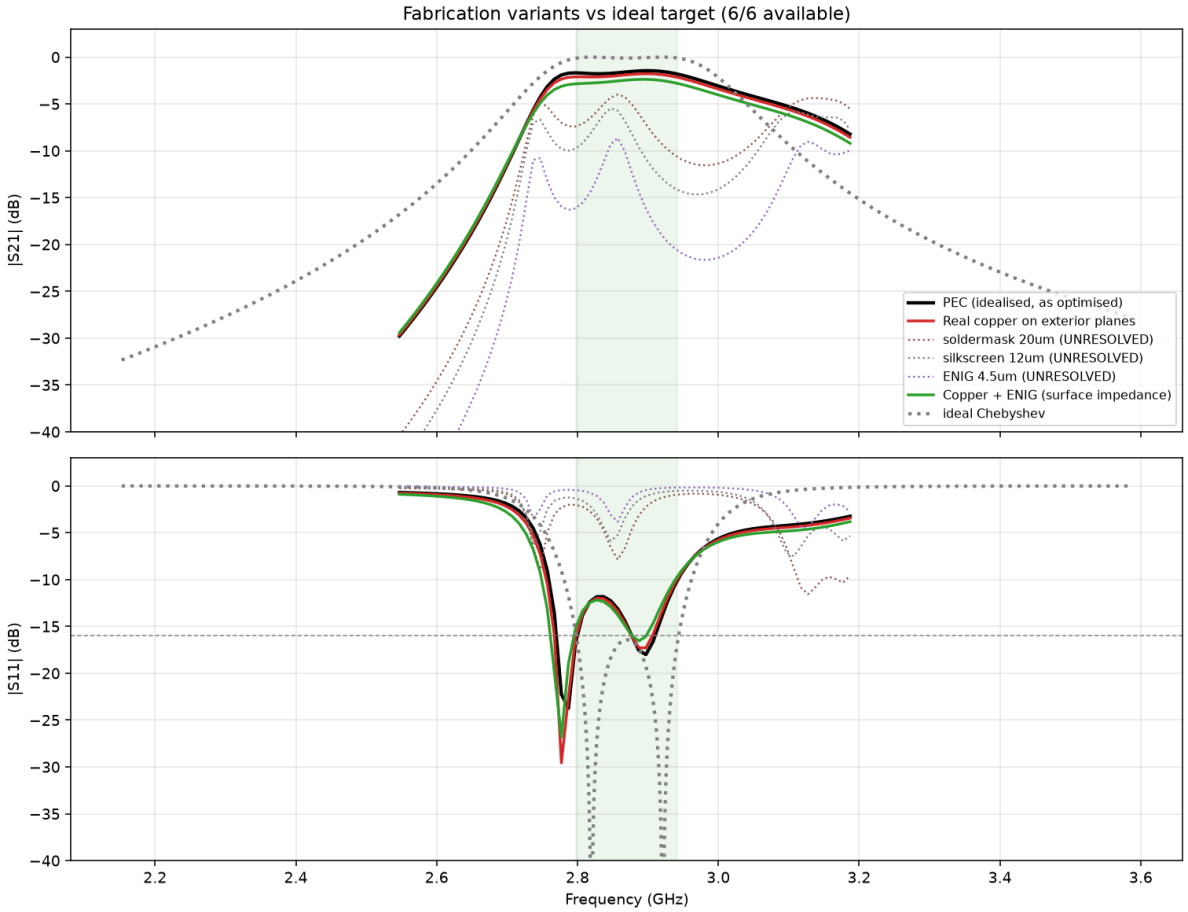


Figure 2: Idealised metal versus real copper (solid) — the real result. The three thin layers (dotted) are shown only to record why they are excluded.

What is true right now

Solid

- The geometry above is a **local optimum for return loss** under the CPWG launch, confirmed by a 30-evaluation search rather than assumed.
- It is **robust to PCB tolerances**: 0.1 mm rounding costs 0.5 dB.
- **Real copper costs 0.32 dB** and does not disturb the response shape.
- The drawn geometry **matches the meshed geometry**. `siw stackup --verify` is no longer a CLI command (removed 2026-09-03); the same comparison now runs as one of the 29 checks in `siw kicad --check` (cascade) or directly via `report.verify_drawing_matches_mesh()` (dualmode) — ground-stitch vias (16), fence vias (78), the 10-vertex feed outline including the CPWG pad taper, and the coplanar gap, all passing.

Not true, and now understood

- **Return loss misses spec**: −10.76 dB against −16 dB.
- **Bandwidth misses spec**: 316.6 MHz against 143.5 MHz.
- These are **one trade, not two faults**. Either can be met alone; neither can be met together on this topology. Meeting the bandwidth costs the match entirely (−1.75 dB at 145.6 MHz).
- **The fabrication stack is unquantified** beyond copper, and cannot be quantified without a mesh this machine cannot build.
- **No board has been ordered.** No CST export, no gerbers, no DRC check.

The one habit that matters

Fifteen silent bugs have now been found in this project's own model. Not one raised an error; every one produced plausible numbers that were wrong. Today added four, and the last of them took three attempts to diagnose:

- an objective term that silently steered every search *away* from the best design, because `optimize()` never forwarded the weights it accepted;
- a "bandwidth curve" that was really the width of one of two disjoint passbands at three of its four points, so inverting it aimed the search at a discontinuity;
- an exterior-face flag hardcoded `True`, correct for a bare board and wrong the moment layers sit above it;
- thin fabrication layers meshing as slivers and destroying the response.

Three lessons, each paid for today:

1. **If a fix does not change the number, the diagnosis was wrong.** Two confident explanations of the −29 dB stack failure were falsified this way, each in one re-run.
2. **After two wrong inferences, stop inferring.** The one-factor ladder settled in four solves what argument had failed to settle twice — and it immediately exonerated the leading suspect.
3. **A correct mechanism aimed at the wrong component is still wrong.** The skin-depth arithmetic was right; it was applied to the vias, which were fine, instead of to the layers, which were not.

Before trusting any result, check it against what physics says it should be. Insertion loss cannot beat a lossless model. A lossy stack cannot cost 27 dB. Pole count must match the filter order. A hole far below cutoff cannot radiate.

Guards now in the code:

- `_driven_config` raises if a mesh domain has no material;
- the mesh builder reports any unclassified exterior face, and raises on an empty mesh instead of handing Palace something valid-looking but empty;
- `band_edges` reports `n_lobes`, so a split response can never again be read as a narrow one;
- `siw scan` excludes split points from its inversion and says why;
- the fabrication runner flags any variant whose loss beats the lossless reference, or is more than 6 dB worse, or shows near-total reflection, or reports zero reflection zeros;
- the drawing-vs-mesh comparison (`report.verify_drawing_matches_mesh` / `verify_cascade_drawing_` folded into `siw kicad --check`) compares the drawing against the meshed geometry vertex by vertex.

Reproducing this

```
# the design
siw design --topology dualmode --N 2 \
    --seed a=57.21,w_iris=15.26,x_iris=28.68,y_wall=8.62,w_ext=16.31 \
    --span 2 --patience 14 --evals 30 --w-bw 0 --device CPU --np 8

# the fabrication study
siw compare-fab-variants --run \
    --params a=57.158,w_iris=15.304,x_iris=28.685,y_wall=9.849,w_ext=16.352 \
    --N 2 --device CPU --np 4

# the trade-off scans
siw scan --topology dualmode --N 2 \
    --params a=57.21,x_iris=28.68,y_wall=8.62,w_ext=15.5 \
    --scan 'w_iris=11,12,13,14' --device CPU --np 8

# artifacts
siw layout --topology dualmode --N 2 \
    --params a=57.158,w_iris=15.304,x_iris=28.685,y_wall=9.849,w_ext=16.352

# fabrication outputs for the topology actually being sent to fab (see the
# 2026-09-04 update above) -- cascade N=3, not this dual-mode design
siw kicad --topology cascade --N 3 \
    --params l_cav1=34.239,l_cav2=35.503,w_iris=14.416,w_ext=13.96,l_taper=15.738,w_ms=1.663
    --launch cpwg
```

Resource limits, learned the hard way: fabrication-stack solves need `--np 4` on this 15 GB machine — eight ranks are killed mid-sweep by the OOM killer — and cost about 32 minutes each.

Building this document

```
cd docs && uv run gen_pdfs.py --force STATE.md   # needs xelatex
pandoc STATE.md -o state.html --standalone --embed-resources --toc
```